

**AN INNOVATIONS AUTOMATED HOSPITAL PATIENT
REGISTRATION AND PRELIMINARY DIAGNOSTIC SYSTEM
USING IOT-BASED SENSORS**



PROJECT WORK

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ABSTRACT

Automated patient registration systems offer a significant advancement in hospital management due to the increasing demand for efficient, accurate, and contactless healthcare solutions. However, managing patient onboarding, identity verification, and preliminary health assessment manually presents a major challenge in high-volume healthcare facilities. This paper explores an intelligent hospital patient registration system integrated with IoT-based sensors for automated vital sign measurement and department allocation. The static analysis on using embedded sensor modules in the healthcare registration process covers body temperature, heart rate, and environmental parameter monitoring. A Raspberry Pi 4 based system is developed with a touchscreen interface to perform real-time data acquisition, Aadhaar-based OTP authentication, and symptom-based triage to determine the appropriate medical department and specialist for consultation.

The IoT-based kiosk system reduces patient waiting time, minimizes manual work, and improves hospital workflow. A Raspberry Pi 4 acts as the central unit, managing sensors via I2C and GPIO. The MLX90614 infrared sensor measures body temperature, while the KY-039 detects heart rate. The BME280 captures humidity and pressure data. Using patient inputs and sensor readings, the system performs preliminary analysis and issues a registration token with a summary, optionally printable.

Keywords: IoT, Patient Registration, Raspberry Pi 4, Aadhaar Authentication, MLX90614, KY-039, BME280, Health Monitoring, Touchscreen Interface, Embedded Systems, Smart Healthcare, Triage Automation.

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LIST OF ABBREVIATIONS

API	: Application Programming Interface
BME280	: Temperature, Humidity and Pressure Sensor
DB	: Database
GPIO	: General Purpose Input Output
GUI	: Graphical User Interface
HR	: Heart Rate
I2C	: Inter-Integrated Circuit
IoT	: Internet of Things
IR	: Infrared
MLX90614	: Infrared Temperature Sensor
OPD	: Outpatient Department
OTP	: One Time Password
PCB	: Printed Circuit Board
PWM	: Pulse Width Modulation
UI	: User Interface
VCC	: Voltage Common Collector

CHAPTER 1

INTRODUCTION

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INTRODUCTION

1.1 OVERVIEW OF THE PROPOSED SYSTEM

The healthcare sector is rapidly evolving with the integration of automation and digital technologies to improve efficiency, accuracy, and patient experience. Traditional hospital registration systems are largely manual, leading to long waiting times, human errors, and inefficient patient management[4]. In high-traffic hospitals, this often results in overcrowding, delayed diagnosis, and reduced quality of service[20].

To address these challenges, hospital automation systems are being introduced to streamline the patient registration and initial assessment process. One such solution is the implementation of a self-service patient registration kiosk, which allows patients to independently enter their details, reducing dependency on administrative staff [15]. This approach not only minimizes waiting time but also improves operational efficiency and ensures a smoother workflow within the hospital environment [16].

In addition to basic registration, modern systems are incorporating sensor-based health screening to perform preliminary medical assessments. By integrating sensors such as temperature sensors, heart rate monitors, and environmental sensors, the system can automatically collect vital health parameters in real time [10]. This enables early detection of abnormal conditions and assists in prioritizing patients based on urgency[21].

The proposed system, titled Automated Hospital Patient Registration System, combines a user-friendly graphical interface, Register number authentication with OTP verification, and real-time sensor data acquisition. It further includes an intelligent decision mechanism to allocate patients to appropriate departments and generate tokens for consultation [2]. The complete

workflow and integration of these modules are illustrated in Figure 1.1, which presents the overview of the IoT-based smart healthcare architecture. This integrated approach enhances the overall efficiency, accuracy, and responsiveness of hospital operations [23].

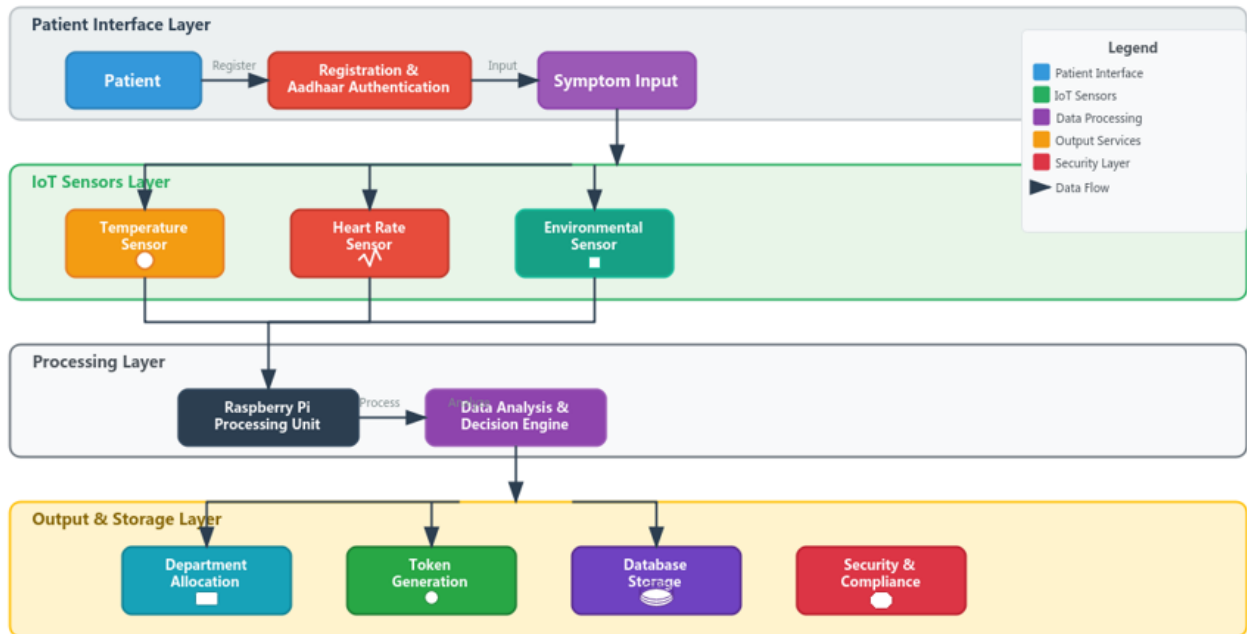


Figure 1.1 Overview of IoT-Based Smart Healthcare Architecture

1.2 IoT IN HEALTHCARE SYSTEMS.

The Internet of Things (IoT) has transformed the healthcare sector by enabling smart, connected medical systems. IoT-based healthcare systems use sensors, communication technologies, and data processing units to monitor patient health in real time. These systems significantly enhance patient monitoring by enabling continuous and remote tracking of vital parameters. They also improve data accuracy by reducing human errors in data collection and recording. Furthermore, IoT supports faster and more efficient clinical decision-making through real-time analytics.

Table 1.1 presents a comparison between Manual and Automated Workflow, highlighting the advantages of IoT-based automation in healthcare systems.

Table 1.1 Comparison of Manual vs Automated Workflow

Parameter	Manual Workflow	Automated Workflow
Registration Method	Paper form / staff entry	Self-service touchscreen
Waiting Time	High	Low
Data Accuracy	Moderate	High
Human Errors	More	Very less
Initial Health Check	Separate process	Integrated sensors
Department Allocation	Manual staff decision	Automatic logic-based
Token Generation	Manual	Automatic
Patient Flow	Slow	Fast

1.2.1 Introduction to IoT-Based Medical Systems

IoT-based medical systems consist of interconnected smart devices and sensors that continuously monitor patient health parameters in real time. Sensors such as temperature sensors and heart rate monitors are used to collect vital physiological data, which is then processed and stored for further analysis and medical decision-making.

In this project, the sensors capture essential patient parameters, and the collected data is processed using an embedded Raspberry Pi-based system. The analyzed results are further utilized for preliminary diagnosis, patient prioritization, and automatic routing to the appropriate hospital department, thereby improving the speed and efficiency of the healthcare workflow.

1.3 SYSTEM DESIGN AND ARCHITECTURE

The proposed system is designed as an integrated smart healthcare kiosk that combines both hardware and software modules to automate patient registration and preliminary diagnosis. The system mainly consists of a Raspberry Pi as the central processing unit, a touchscreen display for user interaction, multiple IoT sensors for measuring temperature, heart rate, and environmental conditions, and a database for secure patient data storage and retrieval.

During operation, the patient first registers through the touchscreen interface, after which register number-based authentication can be performed as an optional verification step. The patient then enters symptoms through the graphical interface, followed by a sensor-based health checkup to capture real-time vital parameters. The collected data is processed and analyzed by the Raspberry Pi, which applies predefined decision logic to determine the appropriate medical department.

Finally, the system performs automatic department allocation and generates a token for consultation, thereby improving hospital workflow efficiency and reducing waiting time. Similar IoT-based healthcare architectures using Raspberry Pi and sensor integration have been widely reported in recent studies. The overall architecture of the proposed system is illustrated in Figure 1.2 .

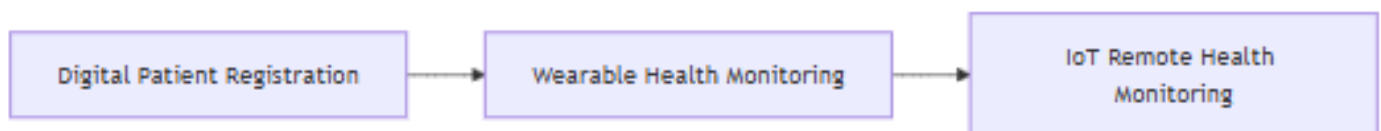


Figure 1.2 Existing System Models

1.4 COMPARISON OF EXISTING REGISTRATION SYSTEMS.

The comparative analysis of patient registration methods clearly highlights the superiority of the proposed IoT-based smart healthcare system over conventional approaches. As presented in Table 1.2 , the manual registration system is affected by slow processing speed, lower data accuracy, lack of health monitoring, and higher patient waiting time, which often leads to inefficiencies in busy hospitals.

The basic digital system offers moderate improvement by reducing paperwork and partially improving data management, but it still lacks real-time diagnostic capabilities and intelligent automation. In contrast, the proposed IoT-based system enables fast patient registration, real-time health parameter monitoring, high automation, and automatic department allocation, thereby significantly improving workflow efficiency and reducing manual intervention.

This comparison clearly demonstrates that the proposed system is better suited for modern smart hospital environments, where speed, accuracy, and intelligent patient routing are essential for quality healthcare delivery.

Table 1.2 Comparison of Existing Patient Registration Systems.

Feature	Manual System	Basic Digital System	Proposed IoT System
Registration Speed	Slow	Moderate	Fast
Data Accuracy	Low	Moderate	High
Health Monitoring	Not Available	Not Available	Available
Automation Level	Low	Medium	High
Waiting Time	High	Medium	Low
Department Allocation	Manual	Partial	Automatic

1.5 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system offers several advantages over conventional registration methods by integrating automation, IoT technology, and intelligent data processing. It significantly improves hospital workflow, reduces manual effort, and enhances patient experience. The system ensures accurate data collection, efficient patient routing, and real-time health monitoring, making it a reliable and effective solution for modern healthcare environments.

1.5.1 Improved Efficiency and Automation.

The automation of the registration and diagnostic process reduces dependency on hospital staff and minimizes delays. By allowing patients to self-register and undergo initial health assessments, the system streamlines the entire workflow and improves operational efficiency. The integration of hardware and software modules ensures smooth coordination between patient input, sensor data acquisition, and result generation.

1.5.1.1 Reduced Waiting Time

The automated system eliminates long queues by enabling quick registration and immediate health checkup. This significantly reduces patient waiting time and improves service delivery. Since the preliminary diagnosis is carried out during the registration stage itself, patients can be directed quickly to the correct department. This improves hospital throughput and reduces congestion in outpatient areas.

1.5.1.2 Automated Vital Sign Monitoring

The integration of sensors allows automatic measurement of vital parameters such as temperature and heart rate, eliminating the need for separate diagnostic procedures. Real-time acquisition of these parameters improves the accuracy of preliminary health assessment and supports faster clinical decisions. The automated process also reduces the workload of nursing and support staff.

1.5.1.3 Accurate Department Allocation

The system analyses patient symptoms and health data to assign the appropriate medical department, ensuring efficient patient routing and reducing errors. This rule-based allocation minimizes the chances of patients visiting the wrong department and avoids unnecessary delays. As a result, both doctors and patients benefit from a more organized consultation process.

1.5.2 Enhanced Data Accuracy

The system ensures high data accuracy by minimizing manual entry and integrating digital verification methods. This reduces the chances of incorrect patient information and improves data reliability. Since all data is stored digitally, duplication of records and handwriting-related errors are eliminated. The structured database design further supports secure and efficient retrieval of patient history.

1.5.2.1 Register number-Based Authentication

The use of Register number-based authentication ensures secure and accurate patient identification, enabling reliable data storage and retrieval. This verification process prevents duplicate registrations and improves patient record consistency. It also makes the system more suitable for institutional or hospital-specific patient management.

1.5.3 Cost-Effectiveness

The proposed system reduces operational costs by minimizing manual labor and optimizing resource utilization. Its compact and scalable design makes it suitable for deployment in various healthcare facilities, providing a cost-effective solution for modern hospitals. The use of Raspberry Pi and low-cost IoT sensors further reduce hardware expenses compared to conventional kiosk systems. This makes the solution practical for both private and government healthcare centers.

1.6 DISADVANTAGES OF THE PROPOSED SYSTEM

Despite its advantages, the proposed system has certain limitations that must be considered. These include dependency on technology, potential usability issues for certain users, and concerns related to data security and privacy. Addressing these challenges is essential for successful implementation and widespread adoption. Proper maintenance, user training, and robust software updates are necessary to overcome these limitations in real-world deployment.

1.6.1 System Dependencies

The system relies heavily on hardware components and network connectivity for proper functioning. Any failure in these components may affect system performance. Hardware malfunctions such as sensor failure or touchscreen issues can interrupt the registration process. Therefore, preventive maintenance and backup support mechanisms are essential.

1.6.1.1 Network Connectivity Requirement

A stable internet connection is required for authentication and data transfer processes, which may be a limitation in certain areas. In rural hospitals or low-network zones, this may reduce the effectiveness of real-time verification and database synchronization. Offline fallback mechanisms may be considered in future improvements.

1.6.1.2 Sensor Calibration Needs

Sensors must be calibrated regularly to ensure accurate measurements, as improper calibration can lead to incorrect data. Continuous usage may cause sensor drift, affecting temperature or heart-rate readings over time. Regular calibration schedules are therefore necessary for maintaining system reliability.

1.6.2 User Limitations

The system may pose challenges for users who are not familiar with digital interfaces, particularly elderly or differently abled individuals. Lack of digital literacy may lead to hesitation or incorrect input during self-registration. User-friendly GUI design and hospital assistance staff can help overcome this limitation.

1.6.2.1 Elderly and Differently Abled Users

Some users, especially elderly or first-time patients, may need assistance while using the touchscreen interface. This dependency can slightly reduce the overall effectiveness of complete automation. Providing voice guidance, larger interface buttons, and multilingual support can improve accessibility for such users.

1.6.2.2 Hardware vs Software Complexity

The integration of multiple hardware and software components increases system complexity, requiring proper maintenance and technical expertise. Troubleshooting such a system may require knowledge of embedded systems, Python software, and sensor interfacing. This can increase maintenance effort in large-scale deployment.

1.6.3 Privacy and Security Concerns

Handling sensitive patient data requires strong security measures to prevent unauthorized access and data breaches. Since medical information is confidential, secure database encryption and controlled access mechanisms are essential. Failure to implement proper cybersecurity can lead to privacy risks.

1.6.3.1 Biometric Data Handling

The use of digital identity verification involves handling sensitive personal data, which must be securely managed. Proper encryption, role-based access, and secure storage protocols are necessary to protect patient information. Compliance with healthcare data privacy standards should also be considered.

1.6.3.2 OTP Dependency

Authentication relies on OTP verification, which may fail due to network issues or unavailability of mobile devices. Such dependency can create delays during urgent hospital visits. Alternative verification methods such as register number login or hospital ID validation can improve system robustness.

1.7 LITERATURE SURVEY

[1] A. Pantelopoulos and N. G. Bourbakis, “A Survey on Wearable Sensor-Based Systems for Health Monitoring and Prognosis,” IEEE Transactions on Systems, Man, and Cybernetics, 2010.

This paper presents a comprehensive survey on wearable sensor-based healthcare systems used for continuous patient monitoring. The study focuses on temperature, pulse rate, ECG, and motion sensors that collect physiological data in real time. The authors explain how wearable sensors improve patient diagnosis by enabling early detection of abnormal conditions. The paper also discusses data processing, storage, and remote communication techniques for medical applications. Experimental studies show that sensor-based monitoring significantly improves diagnostic speed and patient safety. The findings of this work are highly relevant to the proposed project, particularly for integrating temperature and heart rate sensing modules for preliminary diagnosis.

[2] J. Mohammed, C. Lung, A. Ocneanu, A. Thakral, C. Jones, and A. Adler, “Internet of Things: Remote Patient Monitoring Using Web Services and Cloud Computing,” IEEE, 2014.

This paper proposes an IoT-based remote patient monitoring system that uses sensor nodes, cloud services, and web-based healthcare dashboards. The system continuously captures patient vital parameters and stores them in a cloud server for medical analysis. The authors highlight that real-time data acquisition and web-enabled medical access improve patient management and reduce hospital workload. The study validates the system through simulation and experimental healthcare data. This paper strongly supports the use of IoT architecture in the proposed automated patient registration and health monitoring kiosk.

[3] R. Kumar, M. P. Rajasekaran, and S. Daniel, “Automated Patient Registration System Using Raspberry Pi,” IJERT, 2018.

This paper presents a Raspberry Pi-based automated patient registration system designed to reduce hospital queue time and improve front-desk efficiency. The proposed system uses a touchscreen interface for patient data entry and stores the information in a local database. The study emphasizes that automation reduces manual errors and improves data accuracy. Experimental results show significant reduction in waiting time and faster patient processing. This work directly relates to the proposed project, especially in implementing touchscreen-based self-service registration and token generation.

[4] S. Dhanalakshmi and P. Karthick, “Smart Hospital Queue Management Using IoT,” IJIETE, 2019.

This paper introduces an IoT-enabled hospital queue management system that automates patient token generation and doctor consultation flow. The system uses digital interfaces and logic-based routing to minimize overcrowding in hospitals. The authors explain how automated token generation improves patient movement and reduces administrative workload. Simulation results confirm improved

patient flow efficiency. This concept is closely aligned with the proposed department allocation and consultation token generation module.

[5] Y. Liu, X. Ma, and Y. Shu, “IoT-Based Intelligent Healthcare System for Smart Hospitals,” IEEE Access, 2020.

This paper proposes an intelligent healthcare system integrating patient monitoring, automated registration, and smart hospital management. The design combines sensors, embedded processors, and decision logic for improving hospital workflow. The study highlights how IoT-based automation enhances operational speed, data reliability, and medical service quality.

1.8 OUTCOME OF LITERATURE SURVEY

The literature survey clearly indicates that although significant advancements have been achieved in IoT-based healthcare monitoring, patient registration automation, and embedded medical systems, there remains a considerable lack of fully integrated healthcare kiosks that combine all essential hospital front-desk operations into a single automated framework [18], [22]. Many of the surveyed systems provide only isolated functionalities such as sensor-based monitoring, database storage, or registration interfaces, without offering intelligent interconnection among these modules [10], [15].

The analysis of previous research shows that the incorporation of IoT technology, embedded processing, and automated decision logic can significantly improve healthcare services by reducing waiting time, minimizing human errors, and improving the speed of patient routing [16], [21]. Furthermore, studies confirm that sensor-assisted preliminary diagnosis can help in identifying patient urgency levels and improve the efficiency of hospital department allocation [10], [23].

Based on these findings, the proposed system is designed to overcome the major limitations of existing systems by providing a comprehensive, intelligent,

and user-friendly solution for patient registration and preliminary diagnosis. By integrating touchscreen-based registration, Register number verification, real-time sensor acquisition, Raspberry Pi processing, automated token generation, and smart department allocation, the system offers a practical framework for future IoT-enabled smart hospitals.

The outcome of this literature survey strongly supports the feasibility, relevance, and technical significance of the proposed project in improving healthcare service quality [2], [15], [25].

1.9 EXISTING SYSTEM

The existing hospital registration system is primarily manual or partially digital, requiring patients to provide personal details through forms or basic digital interfaces. These systems do not include real-time health monitoring or intelligent decision-making features. As a result, patients must undergo separate diagnostic procedures, leading to increased waiting time and inefficient workflow.

Additionally, manual systems are prone to data entry errors and require significant human intervention. Even digital systems lack integration with advanced technologies such as IoT and automated analysis, making them less effective in handling large patient volumes efficiently.

1.10 PROBLEM IDENTIFICATION

The current healthcare registration systems face several challenges, including long waiting times, inefficient patient routing, and lack of integration between registration and diagnostic processes. Manual data entry increases the risk of errors, while the absence of real-time health monitoring delays diagnosis.

Furthermore, the lack of intelligent systems for department allocation results in improper patient routing, causing overcrowding in certain departments. These issues highlight the need for an automated and integrated solution to improve healthcare efficiency.

1.10.1 SOLUTIONS

To overcome the identified problems, the proposed system introduces automation, IoT integration, and intelligent data processing. The system enables self-registration, real-time health monitoring, and automatic department allocation. It reduces manual intervention, improves data accuracy, and enhances overall workflow efficiency.

By integrating multiple functionalities into a single platform, the system provides a comprehensive solution that addresses the limitations of existing healthcare registration systems.

1.11 PROPOSED SYSTEM

The proposed system is an intelligent IoT-based healthcare kiosk designed to automate patient registration and preliminary diagnosis. It integrates a touchscreen interface, embedded sensors, and data processing capabilities to provide a seamless and efficient user experience. Patients can register, input symptoms, and undergo basic health checks in a single step.

The system processes the collected data and generates a unique token along with a patient summary, including health parameters and department allocation. This reduces waiting time, improves accuracy, and enhances hospital workflow. Overall, the proposed system represents a significant advancement in healthcare automation and contributes to the development of smart healthcare infrastructure.

CHAPTER 2

BLOCK DIAGRAM AND DESCRIPTION

CHAPTER 2

BLOCK DIAGRAM AND DESCRIPTION

2.1 BLOCK DIAGRAM OF THE SYSTEM

The proposed system is an integrated IoT-based healthcare kiosk for automated patient registration and preliminary diagnosis. It includes a Raspberry Pi, touchscreen interface, sensor modules, and output components for real-time data processing. The Raspberry Pi acts as the central controller, coordinating communication between all hardware and software modules. The complete functional structure is illustrated in Figure 2.1 Block Diagram.

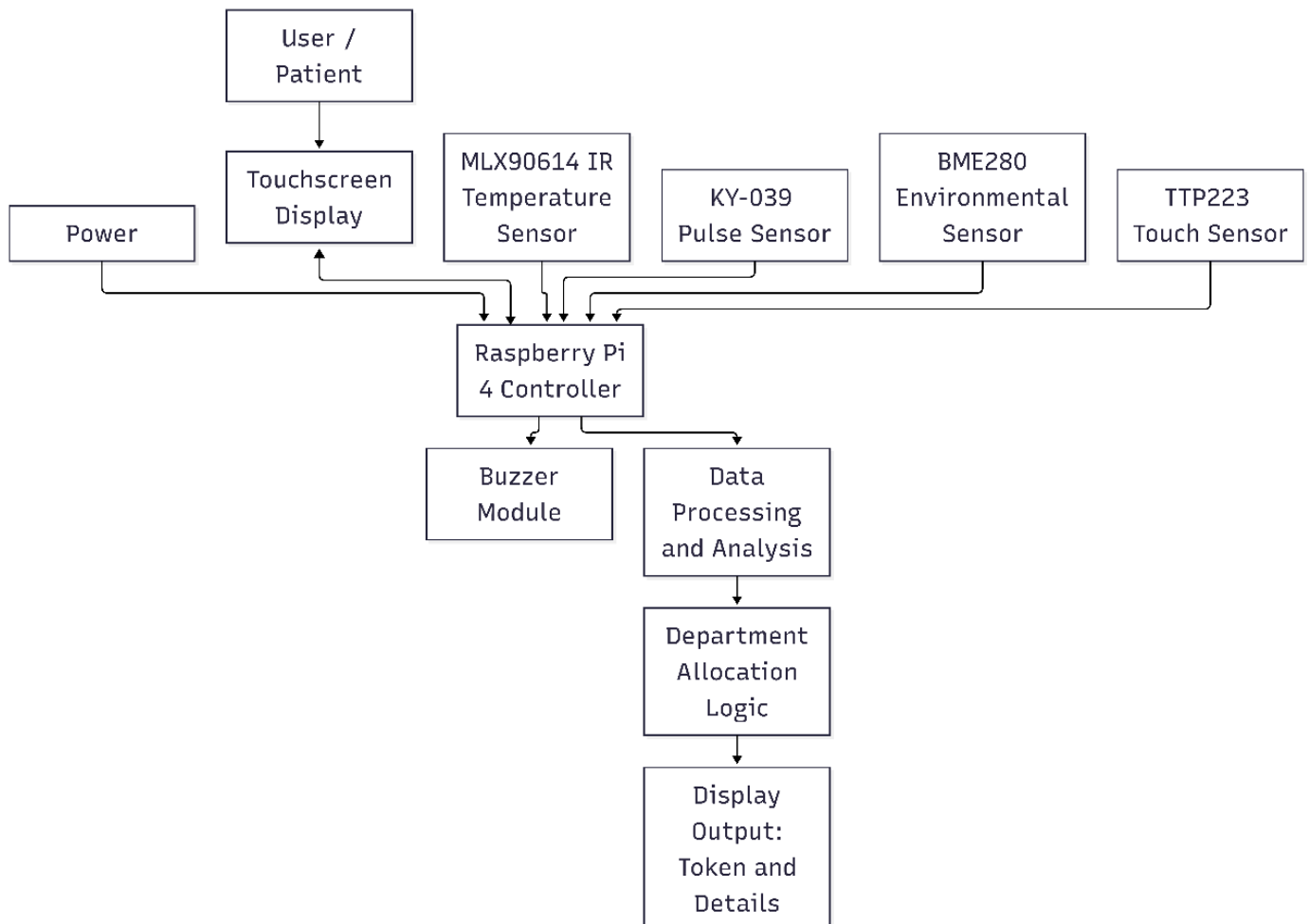


Figure 2.1 Block Diagram

The proposed system follows a structured and sequential workflow to ensure smooth healthcare kiosk operation. Initially, the patient interacts with the touchscreen interface to enter personal details and symptoms. Once the registration process is completed, the integrated IoT sensors capture essential health parameters such as body temperature and heart rate.

The collected patient and sensor data are then processed and analyzed by the Raspberry Pi, where the embedded decision logic determines the appropriate medical department based on the symptoms and measured values[15]. Following this analysis, the department is allocated automatically, and the system generates both a consultation token and a patient summary report for further medical review.

This block-level system architecture ensures efficient workflow execution, reduces manual intervention, minimizes waiting time, and significantly improves healthcare service delivery [25].

2.2 RASPBERRY PI 4 MODEL B

The Raspberry Pi 4 Model B serves as the main processing unit of the system. It is a compact and powerful single-board computer capable of handling multiple operations simultaneously. It controls the graphical user interface, collects sensor data, processes inputs, and generates outputs.

The Raspberry Pi enables seamless integration of hardware and software components, making it suitable for real-time healthcare applications.

2.3 TOUCHSCREEN DISPLAY

The touchscreen display serves as the primary human–machine interface (HMI) between the patient and the proposed smart healthcare system, enabling direct interaction without the need for external input devices such as a keyboard or mouse. It is responsible for displaying patient registration forms, allowing symptom input, showing step-by-step instructions and feedback messages, and finally presenting the generated token along with the patient summary details.

Through the touchscreen, users can easily navigate the complete registration and preliminary diagnosis workflow in a guided manner.

The graphical user interface (GUI) is designed to be simple, intuitive, and user-friendly, with clear buttons, readable text, and guided prompts to ensure accessibility for patients of different age groups and technical backgrounds. This improves usability, reduces dependency on hospital staff, and supports smooth self-service operation within the healthcare kiosk[3].

2.4 MLX90614 IR TEMPERATURE SENSOR

The MLX90614 is a non-contact infrared temperature sensor used to measure body temperature. It detects infrared radiation emitted by the human body and converts it into temperature readings.

$$P = \sigma \epsilon A T^4 \dots\dots\dots (1)$$

The sensor offers several important advantages that make it highly suitable for healthcare applications. Since it performs contactless temperature measurement, it ensures safe and hygienic operation, which is highly important in hospital environments to reduce the risk of cross-contamination. The sensor provides high accuracy of approximately $\pm 0.5^\circ\text{C}$ with fine measurement resolution, enabling reliable fever detection and preliminary diagnosis.

In addition, its fast response time and digital I2C communication support allow quick integration with the Raspberry Pi for real-time monitoring. These features make the MLX90614 an effective component for automated smart healthcare kiosks. The sensor communicates with the Raspberry Pi using the I2C protocol and plays a crucial role in preliminary diagnosis[9].

2.5 KY-039 PULSE SENSOR

The KY-039 pulse sensor is used to measure the heart rate of the patient. It operates based on light absorption variations caused by blood flow.

$$\text{BPM} = \frac{60}{T} \dots \dots \dots (2)$$

T = time interval between pulses (seconds)(3)

$$\text{Heart Rate} = \frac{\text{Number of pulses}}{\text{Time}} \times 60$$

The KY-039 heart rate sensor works based on the optical pulse detection method, where an infrared LED and phototransistor are used to sense blood flow variations in the finger. For accurate measurement, the sensor requires proper finger placement, ensuring stable light transmission and reflection through the fingertip.

It provides real-time heart rate data, which helps in continuously monitoring the patient's pulse condition during the preliminary screening stage. The collected signal data is further processed by the Raspberry Pi to determine the patient's heart rate in beats per minute (BPM), making it an important component in the proposed smart healthcare system.

2.6 BME280 ENVIRONMENTAL SENSOR

The BME280 sensor measures environmental parameters such as temperature, humidity, and atmospheric pressure. These values help in providing additional context for patient health monitoring.

$$P = \frac{F}{A} \dots \dots \dots (4)$$

Features:

- High accuracy measurement
- Low power consumption
- I2C communication support

The sensor enhances the system by providing environmental data that can influence patient conditions.

2.7 TOUCH SENSOR (TTP223)

The TTP223 touch sensor is used as an additional input device. It detects touch through capacitive sensing and triggers system actions.

Functions:

- Start measurement process
- Confirm inputs
- Improve user interaction

It provides a simple and reliable way to interact with the system.

2.8 BUZZER MODULE

The buzzer module provides audio feedback during system operation, helping users navigate through different stages of the healthcare kiosk process. It generates distinct alert tones to indicate events such as successful data input, error notifications, and completion of sensor measurements.

These audio indications improve the overall usability, user guidance, and interaction experience, especially for first-time users and elderly patients. By providing immediate feedback, the buzzer ensures smooth and reliable system operation, reduces user confusion, and supports efficient workflow execution within the smart healthcare kiosk[13].

2.9 POWER SUPPLY

The power supply unit provides stable electrical power to all the components of the proposed smart healthcare system, ensuring smooth and uninterrupted operation. A regulated 5V DC supply is used to safely power the Raspberry Pi, sensors, touchscreen display, and supporting modules, thereby maintaining consistent performance throughout the patient registration and diagnosis process.

The regulated supply helps prevent issues caused by voltage fluctuations, sudden drops, or overload conditions, which could otherwise affect sensor accuracy and system reliability. Therefore, the power supply unit is an essential part of the system architecture, contributing significantly to safe, stable, and efficient overall operation.

2.9.1 Use and Function of Power Supply

The power supply unit plays a critical role in ensuring the stable and reliable operation of the proposed smart healthcare system. It provides a regulated and constant voltage to the Raspberry Pi, ensuring uninterrupted processing and smooth execution of the patient registration and diagnosis workflow. In addition, the power supply supports reliable operation of connected sensors and peripheral modules, allowing accurate acquisition of temperature, heart rate, and other medical parameters[24].

It also offers protection against voltage fluctuations, surges, and sudden drops, thereby safeguarding sensitive electronic components from malfunction or damage. As listed in Table 2.1 Bill of Materials – Hardware Components below, the selected regulated 5V, 3A power supply module is suitable for driving the Raspberry Pi, touchscreen display, and all interfaced sensors.

Hence, proper power management is essential for maintaining overall system stability, measurement accuracy, and long-term performance of the healthcare kiosk.

Table 2.1 Bill of Materials – Hardware Components

Sl. No	Component Name	Specification / Description	Quantity
1	Raspberry Pi 4 Model B	4GB/8GB RAM, main processing unit	1
2	MicroSD Card	32GB or higher storage	1
3	Touchscreen Display	7-inch HDMI/DSI compatible	1
4	Power Adapter	5V, 3A regulated supply	1
5	BME280 Sensor	Temperature, humidity, pressure sensor	1
6	MLX90614 Sensor	Infrared temperature sensor	1
7	KY-039 Pulse Sensor	Heart rate sensor	1
8	TTP223 Touch Sensor	Capacitive touch module	1
9	Buzzer Module	Audio alert device	1
10	Jumper Wires	Connecting wires	As req.
11	Breadboard/PCB	Circuit connections	1
12	Enclosure	Kiosk structure	1

CHAPTER 3

CIRCUIT DIAGRAM AND DESCRIPTION

CHAPTER 3

CIRCUIT DIAGRAM AND DESCRIPTION

3.1 CIRCUIT DIAGRAM OF THE PROPOSED SYSTEM

The circuit diagram illustrates the interconnection of all hardware components used in the proposed smart healthcare system, with the Raspberry Pi serving as the central controller that interfaces with the sensors, touchscreen display, buzzer, and other output modules.

The design ensures a proper voltage supply, reliable communication between components, and safe as well as stable system operation for continuous healthcare monitoring and patient registration tasks. The complete hardware wiring and component connections are clearly shown in Figure 3.1 Circuit Diagram.

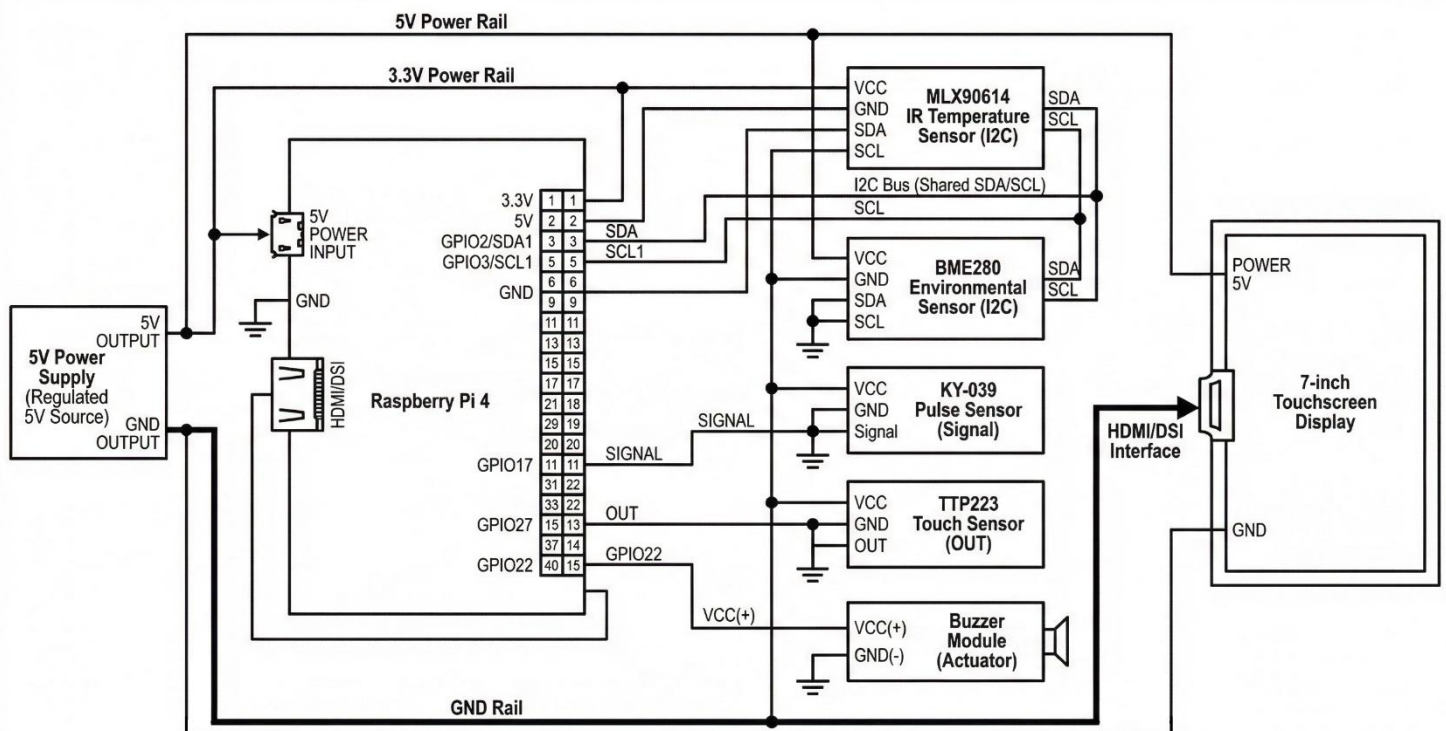


Figure 3.1 Circuit Diagram

3.1.1 Circuit Diagram Description

The Raspberry Pi acts as the central hub, connecting to sensors and peripherals. The MLX90614 and BME280 sensors are connected via the I2C interface, while the pulse sensor and buzzer are connected through GPIO pins. The touchscreen display is connected using HDMI or DSI interface.

Proper grounding and voltage levels are maintained to ensure safe operation. The circuit design focuses on simplicity, reliability, and scalability.

3.2 POWER SUPPLY

The power supply circuit is designed to provide a stable regulated 5V output required for the Raspberry Pi and all connected hardware modules, ensuring uninterrupted system performance. It incorporates voltage regulation and basic protection mechanisms to safeguard the controller and sensors from fluctuations, surges, and sudden drops. This stable supply is essential for maintaining safe, reliable, and continuous operation of the smart healthcare kiosk.

3.2.1 Block Diagram of Power Supply

The power supply block consists of an AC input stage, rectifier, filter, and voltage regulator, which works together to convert AC power into a stable DC output suitable for the proposed system.

This regulated supply ensures reliable operation of the Raspberry Pi, sensors, and display modules without interruption. Proper filtering reduces ripples and improves the quality of the output voltage for safe hardware performance. The complete design is illustrated in Figure 3.2 Power Supply Circuit below.

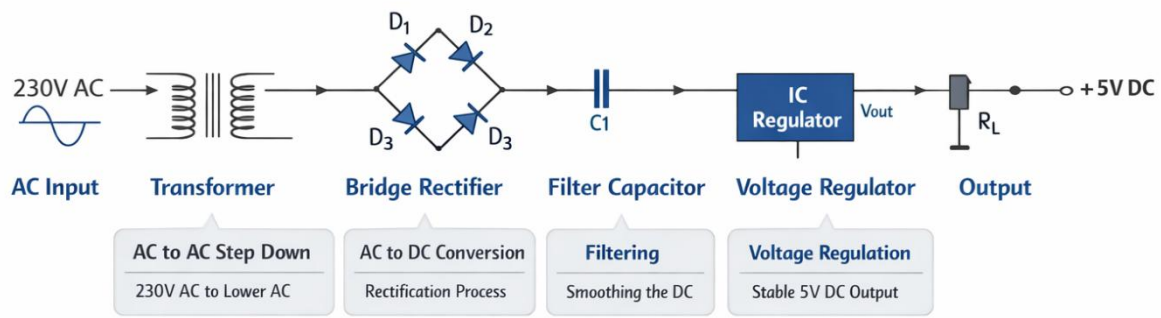


Figure 3.2 Power Supply Circuit

3.2.2 Voltage Regulation

Voltage regulation is achieved using a regulator circuit that maintains a constant output voltage despite variations in the input supply, ensuring stable performance of the proposed smart healthcare system. This is essential because the Raspberry Pi, sensors, and touchscreen modules require a steady 5V supply for accurate operation.

The regulator compensates for sudden voltage rises, drops, or ripple components that may occur from the rectifier and filtering stages. As a result, it protects sensitive electronic components from malfunction, data errors, or permanent damage caused by voltage fluctuations. Proper voltage regulation also improves overall reliability, measurement accuracy, and long-term lifespan of the healthcare kiosk hardware.

3.3 RASPBERRY PI

The Raspberry Pi is interfaced with sensors and peripheral modules through multiple communication protocols such as GPIO, I2C, SPI, and HDMI, enabling efficient data exchange within the proposed system. It receives real-time inputs from temperature, pulse, and environmental sensors, processes the collected data, and controls system operations such as display output, buzzer alerts, and token generation.

The Raspberry Pi also manages communication with the SQLite database and decision engine for department allocation. This centralized control ensures smooth coordination between hardware and software modules for reliable healthcare automation. The detailed GPIO and interface layout is shown in Figure 3.3 Raspberry Pi Pin Diagram.

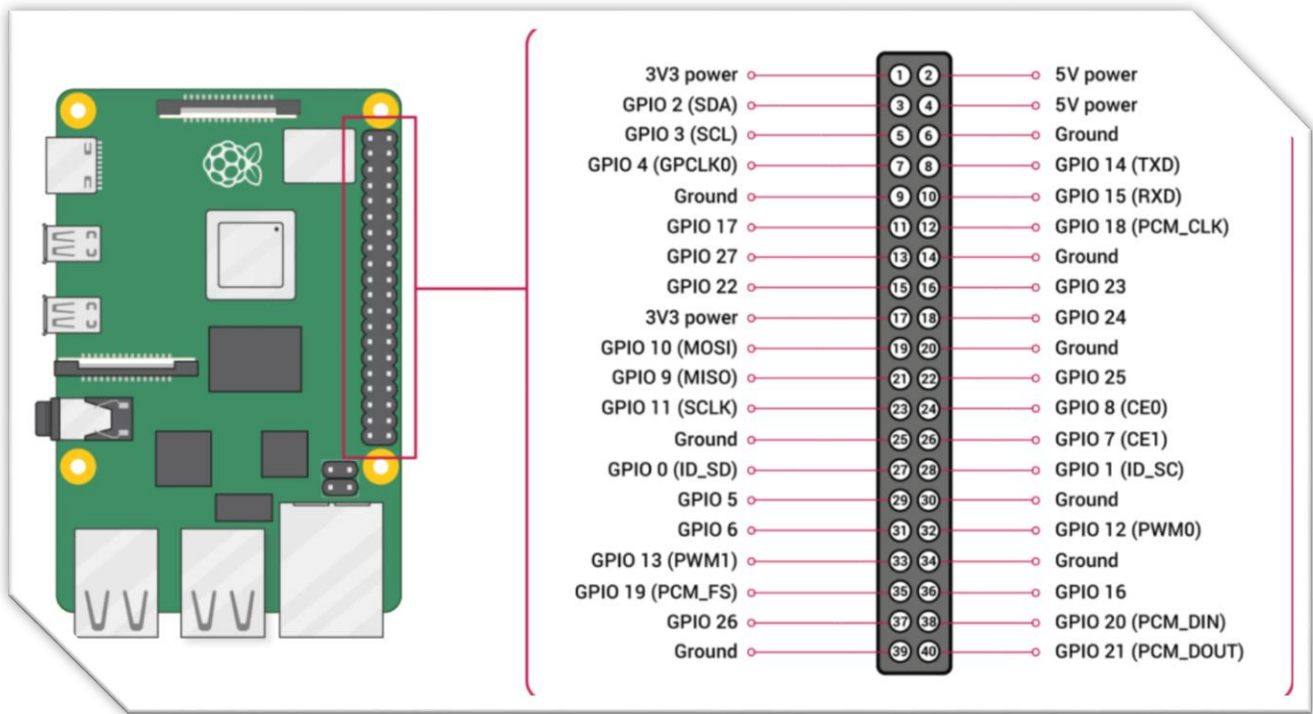


Figure 3.3 Raspberry Pi Pin Diagram

3.3.1 GPIO Interface

The GPIO pins of the Raspberry Pi are used to interface external components such as the pulse sensor and buzzer module. These pins allow the Raspberry Pi to send and receive digital signals, enabling efficient control, signal triggering, and real-time data acquisition from connected devices. Through these GPIO connections, the system can process sensor outputs and activate alert modules based on the patient's health status.

The detailed pin functions and connections used in the proposed system are presented in Table 3.1, which provides the Raspberry Pi 4 GPIO pin description. This GPIO-based interfacing plays an important role in ensuring reliable hardware communication and smooth system operation.

Table 3.1 Raspberry Pi 4 GPIO Pin Description

Pin No	Name	Function
1	3.3V	Power Supply
2	5V	Power Supply
3	SDA	I2C Data
5	SCL	I2C Clock
6	GND	Ground
7	GPIO4	General Input/Output

3.3.2 I2C Communication

I2C communication is used to interface sensors like MLX90614 and BME280. It allows multiple devices to communicate using a two-wire protocol, simplifying circuit design and reducing wiring complexity.

3.3.3 Processing Capability

The Raspberry Pi processes sensor data, executes algorithms, and manages the graphical user interface. Its high processing capability ensures smooth and efficient system operation.

3.4 MLX90614 IR SENSOR

The MLX90614 infrared temperature sensor is connected to the Raspberry Pi through the I2C communication interface, enabling accurate and real-time body temperature measurement in the proposed healthcare system. It requires proper address configuration and stable SDA/SCL communication to ensure reliable data transfer and precise sensing performance.

The sensor works by detecting infrared radiation emitted from the human body and converting it into corresponding electrical signals, which are then processed by the Raspberry Pi to obtain temperature readings for preliminary health assessment. The sensor interfacing and pin connections are illustrated in

Figure 3.4 MLX90614 Sensor Diagram , while the detailed pin functions are listed in Table 3.2 MLX90614 IR Sensor Pin Description.

Pinout of MLX90614

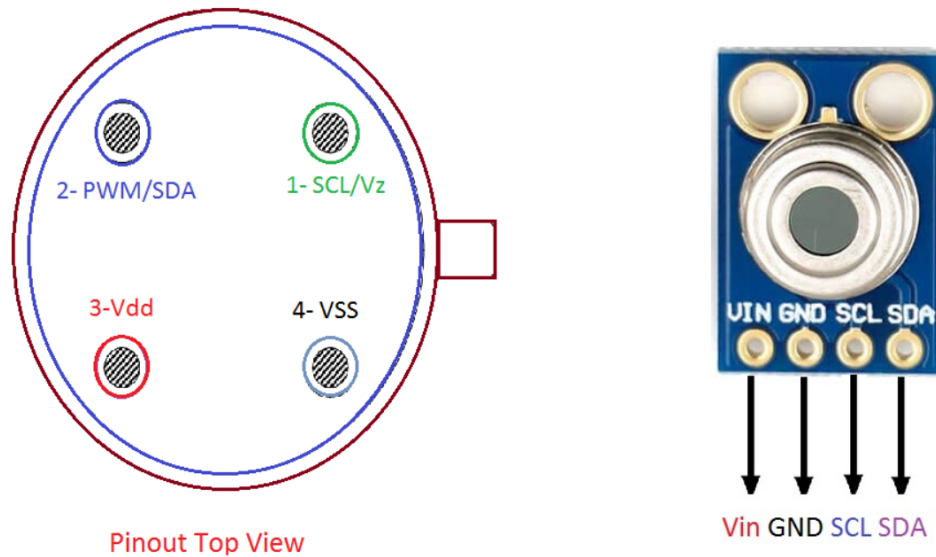


Figure 3.4 MLX90614 Sensor Diagram

Table 3.2 MLX90614 IR Sensor Pin Description

Pin	Name	Function
1	VCC	Power Supply
2	GND	Ground
3	SDA	Data Line
4	SCL	Clock Line

3.5 KY-039 PULSE SENSOR

The pulse sensor is connected to the GPIO pins of the Raspberry Pi. It requires analog signal processing to extract heart rate information.

The sensor detects blood flow variations and converts them into electrical signals. These signals are filtered and processed to calculate heart rate accurately.

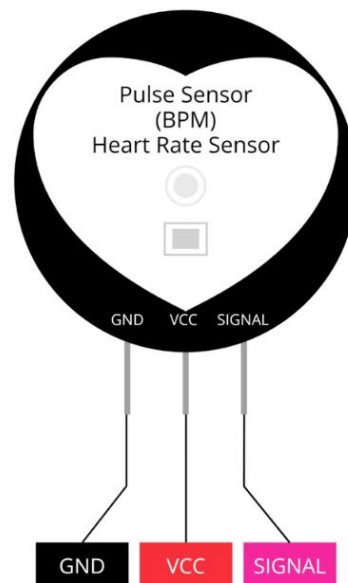


Figure 3.5 Pulse Sensor Diagram.

3.6 BME280 SENSOR

The BME280 sensor is interfaced with the Raspberry Pi using the I2C communication protocol, enabling reliable and low-power digital data exchange. It is used to measure important environmental parameters such as temperature, humidity, and atmospheric pressure, which can support surrounding condition monitoring within the healthcare kiosk environment.

The sensor provides high-accuracy digital output, fast response characteristics, and stable long-term performance, making it suitable for real-time embedded healthcare applications. In the proposed system, the environmental data obtained from the BME280 can be used to monitor room conditions and support better patient comfort and sensor calibration reliability.

3.6.1 Pin Diagram

The BME280 sensor includes pins for power supply, ground, and I2C communication lines (SDA and SCL), which are essential for reliable interfacing with the Raspberry Pi. Proper connection of these pins ensures accurate environmental data transmission, including temperature, humidity, and pressure readings, without communication errors.

Stable wiring and correct voltage levels are important for maintaining consistent sensor performance in the healthcare kiosk. The pin connections and interfacing layout are illustrated in Figure 3.6 BME280 Diagram.

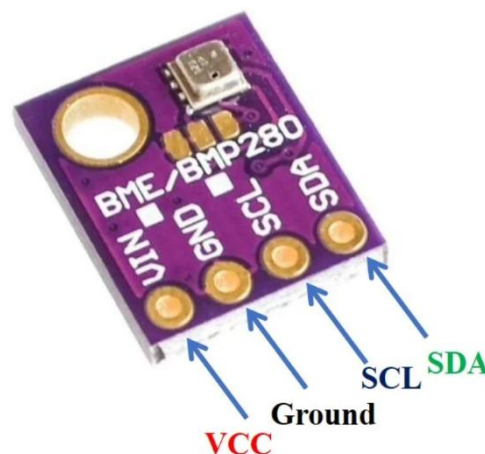


Figure 3.6 BME280 Diagram

3.6.2 Pin Description

Each pin has a specific function, such as power input, data communication, and clock signal. Correct pin configuration ensures reliable sensor operation.

Table 3.3 BME280 Sensor Pin Description

Pin	Name	Function
1	VCC	Power Supply
2	GND	Ground
3	SDA	Data Line
4	SCL	Clock Line

3.7 TTP223 TOUCH SENSOR

The TTP223 touch sensor is connected to the Raspberry Pi through GPIO pins, where it detects human touch input and generates corresponding digital signals for system control.

When the user touches the sensor surface, the module sends an input signal to the Raspberry Pi, which is used to trigger functions such as start, confirmation, or navigation actions within the healthcare kiosk.

This touch-based interface enhances user interaction, accessibility, and ease of operation, especially for patients who require a simple and responsive input method. The complete interfacing and module layout is shown in Figure 3.7 Touch Sensor Module.



Figure 3.7 Touch Sensor Module

3.8 BUZZER MODULE

The buzzer module is connected to one of the GPIO pins of the Raspberry Pi and is controlled programmatically to generate audio indications during system operation. It produces sound signals to represent different system states, such as successful input confirmation, warning alerts, error notifications, and completion of sensor measurements.

These audible cues help guide the user through each stage of the healthcare kiosk process, making the interaction more intuitive and user-friendly. Thus, it plays an important role in enhancing usability, system guidance, and smooth workflow execution within the proposed smart healthcare system.

3.9 TOUCHSCREEN DISPLAY

The touchscreen display is connected to the Raspberry Pi using either the HDMI or DSI interface and serves as the primary human–machine interface for patient interaction within the proposed healthcare kiosk. It displays system menus, registration forms, symptom input fields, instructions, and final results, thereby providing a complete and user-friendly interactive experience.

The display ensures smooth navigation through each stage of the registration and preliminary diagnosis workflow. The touchscreen module and its interfacing arrangement are illustrated in Figure 3.8 Touchscreen Display.

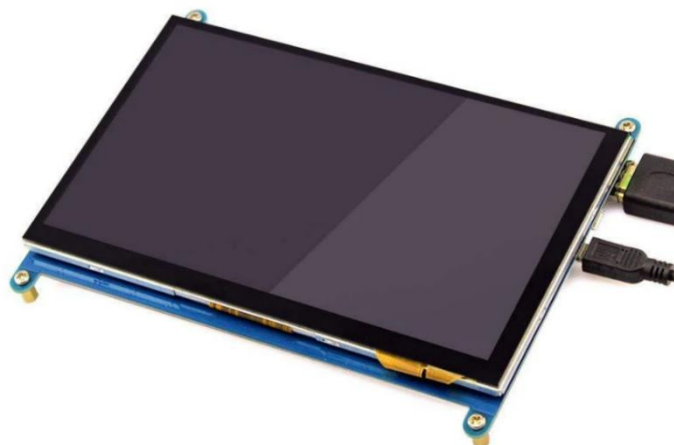


Figure 3.8 Touchscreen Display

CHAPTER 4
TECHNOLOGY USED

CHAPTER 4

TECHNOLOGY USED

4.1 BACKGROUND KNOWLEDGE

The proposed Automated Hospital Patient Registration System is developed using a combination of embedded systems, Internet of Things (IoT), and software technologies to automate patient registration and preliminary diagnosis processes. Embedded systems play a crucial role in controlling hardware components such as temperature sensors, heart rate sensors, buzzer modules, touchscreen interfaces, and GPIO-based peripherals. In this project, the Raspberry Pi serves as the core embedded platform, enabling real-time data acquisition, processing, and system control .

IoT technology enables seamless communication between sensors and the processing unit, allowing continuous monitoring of patient health parameters such as body temperature and heart rate. These technologies are widely adopted in modern healthcare systems to improve speed, accuracy, accessibility, and operational efficiency [2], [25].

In addition to hardware integration, knowledge of software development, database management, and graphical user interface (GUI) design is essential for implementing a complete and user-friendly healthcare kiosk. The integration of these technologies results in a smart healthcare solution capable of reducing manual workload, minimizing waiting time, and improving overall patient experience, making it highly suitable for next-generation hospital automation systems [15], [23].

4.2 RELATED WORKS

Several research studies have explored the application of IoT, embedded systems, and automation technologies in healthcare environments, particularly in the areas of digital patient registration, remote health monitoring, and smart

hospital management [2], [15], [25]. Existing systems include touchscreen-based patient registration platforms, wearable health monitoring devices, and cloud-assisted medical data acquisition systems. These systems have demonstrated significant improvements in patient data management, monitoring accuracy, and hospital workflow efficiency, but most of them still lack complete integration between the various functional modules [10], [23].



Figure:4.1 Hardware connections

Many hospital kiosk systems focus only on the registration process without incorporating real-time diagnostic capabilities such as temperature and heart-rate collecting patient data continuously, but they often do not include intelligent decision-making, symptom analysis, or automated department allocation mechanisms [16], [21]. This separation of functionalities leads to fragmented workflows, repeated data entry, delayed patient routing, and reduced operational efficiency in busy healthcare environments.

The proposed system addresses these limitations by combining patient registration, Register number-based verification, sensor-based health monitoring, intelligent data analysis, automatic department allocation, and token generation into a single unified smart healthcare platform.

This level of integration significantly enhances system efficiency, reduces patient waiting time, minimizes manual intervention, and improves the overall quality and responsiveness of healthcare services [15], [25].

4.3 SOFTWARE IMPLEMENTATION

The software implementation of the proposed smart healthcare system is carried out using Python due to its simplicity, flexibility, and extensive support for libraries related to GUI development, sensor interfacing, and database management. The software is responsible for handling user interaction, acquiring and processing sensor data, performing decision-based analysis, and generating outputs such as patient summaries and consultation tokens.

The software architecture ensures smooth communication between hardware components and Raspberry Pi, while efficiently managing real-time data processing, decision logic, and output generation, thereby supporting the reliable operation of the proposed healthcare kiosk [15].

4.4 ALGORITHM

The proposed system follows a structured step-by-step algorithm to automate patient registration and preliminary diagnosis efficiently. The process begins with the graphical user interface (GUI), where the patient enters personal details and symptoms.

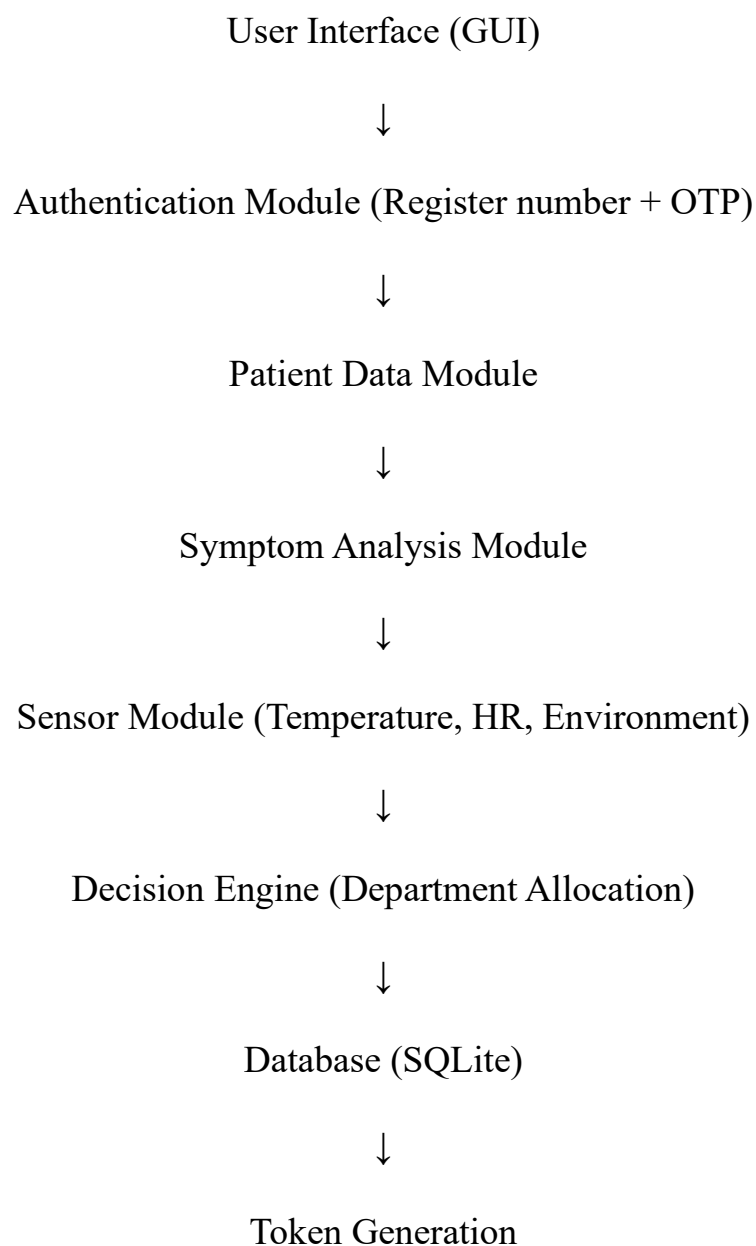
The entered information is then verified through the authentication module using register number and OTP validation to ensure secure and accurate patient identification. After successful verification, the patient data module stores the collected details, and the symptom analysis module evaluates the entered symptoms.

Simultaneously, the sensor module acquires real-time physiological parameters such as body temperature, heart rate, and environmental conditions.

The collected data is processed by the decision engine, which applies predefined logic to allocate the appropriate medical department.

The processed information is then stored in the SQLite database for future reference, after which the system performs automatic token generation for consultation. This structured algorithm ensures fast processing, reduced manual intervention, and improved hospital workflow efficiency.

Algorithm Steps

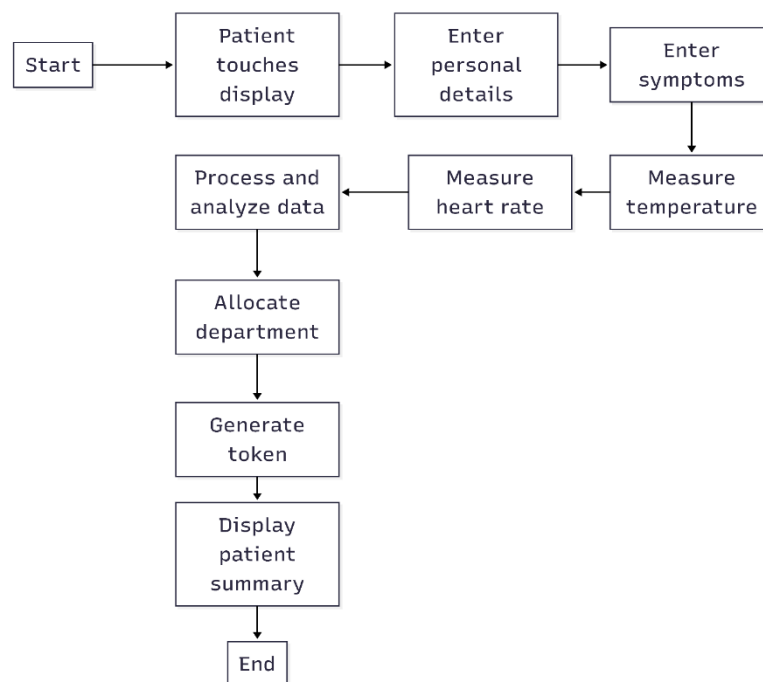


4.5 PROGRAMMING IN FLOWCHART

The system flow chart represents the sequential execution of the software and hardware workflow steps in the proposed healthcare kiosk. It begins with user input through the touchscreen interface, where the patient enters personal details and symptoms.

This is followed by authentication and sensor-based data collection, during which the system captures body temperature, heart rate, and other required parameters. The collected data is then processed using predefined decision logic and analysis algorithms to determine the most appropriate medical department based on the patient's condition.

Finally, the system generates a consultation token and displays the patient summary report for further hospital processing. The complete step-by-step execution sequence is illustrated in Figure 4.1, which shows the system flowchart of the proposed software implementation. This structured workflow ensures efficient, accurate, and reliable system operation, while minimizing manual intervention and improving patient service quality [15], [25].



4.2 System Flowchart – Patient Registration Process

4.6 SYMPTOM ANALYSIS AND DEPARTMENT ALLOCATION LOGIC

The system includes an intelligent module for analysing patient symptoms and sensor data to determine the appropriate medical department. This logic is based on predefined rules and conditions that map symptoms to specific departments.

Working Principle

The proposed system operates based on an intelligent rule-based decision workflow for patient routing and preliminary diagnosis. Initially, the system collects the symptoms entered by the patient through the touchscreen interface. In parallel, sensor data such as body temperature and heart rate are acquired and analysed using the connected IoT modules. These patient inputs and measured health parameters are then evaluated using predefined medical decision rules to determine the most suitable hospital department.

For example, if the system detects fever along with high body temperature, the patient is routed to General Medicine, whereas chest pain combined with an abnormal heart rate leads to Cardiology, and breathing-related issues are directed to Pulmonology.

This logic-based approach offers several important advantages. It significantly reduces manual intervention by hospital staff, improves the accuracy of patient routing, and saves valuable time for both patients and doctors. In addition, it enhances the overall operational efficiency of the hospital by reducing queue congestion and improving the speed of consultation flow.

The implementation of this intelligent decision logic ensures that patients are directed to the correct department quickly and efficiently, thereby improving healthcare service delivery and patient experience [16], [25].

CHAPTER 5

TEST RESULT AND DISCUSSIONS

CHAPTER 5

TEST RESULT AND DISCUSSIONS

5.1 SOFTWARE SPECIFICATION

The proposed system is developed using a combination of software tools and technologies to ensure efficient operation and user interaction. The software is designed to handle multiple tasks such as user input processing, sensor data acquisition, data analysis, and result display.

The system runs on a Raspberry Pi platform with a Linux-based operating system. Python is used as the primary programming language due to its simplicity and extensive support for hardware interfacing and GUI development. The software is modular in nature, allowing easy maintenance and scalability.

Key software requirements include:

- Operating System: Raspberry Pi OS
- Programming Language: Python
- GUI Framework: Tkinter
- Communication Protocols: I2C and GPIO libraries
- Database: Local storage for patient records

The software ensures smooth interaction between hardware and user interface, enabling real-time processing and reliable system performance.

5.2 PYTHON

Python is used as the core programming language for implementing the system due to its readability, flexibility, and wide range of libraries. It enables easy integration with hardware components and supports rapid development of applications.

5.2.1 Overview

Python is a high-level interpreted programming language widely used in embedded systems, automation, and IoT applications due to its simplicity, flexibility, and extensive library support. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming, making it highly suitable for developing smart healthcare solutions.

In the proposed project, Python is used for interfacing sensors with the Raspberry Pi, processing patient information, implementing the system decision logic, and developing the graphical user interface for patient interaction. Its rich ecosystem of libraries enables smooth communication between hardware and software modules, ensuring efficient real-time data acquisition, analysis, and output generation within the smart healthcare kiosk.

5.2.2 Libraries Used

Several Python libraries are utilized to implement the different functional modules of the proposed smart healthcare system. The Tkinter library is used for developing the graphical user interface (GUI), enabling interactive patient registration and symptom entry through the touchscreen display. The smbus library is employed for I2C communication with sensors such as the MLX90614 and BME280, ensuring smooth real-time data acquisition.

The RPi.GPIO library is used for controlling Raspberry Pi GPIO pins, allowing interfacing with modules such as the pulse sensor, touch sensor, and buzzer. In addition, the time library is used to manage timing operations and delay functions, which are essential during sensor readings and output generation. For patient data storage and retrieval, the sqlite3 library is used for database management, enabling efficient handling of patient records and generated tokens.

Together, these libraries ensure seamless communication between software and hardware components, resulting in efficient system performance and reliable healthcare automation.

5.2.3 GUI Development

Tkinter is used to develop the graphical user interface (GUI) of the proposed system, providing a simple, lightweight, and effective platform for creating interactive healthcare applications on Raspberry Pi. It enables the design of patient-friendly screens for registration, symptom input, sensor status display, and token generation, ensuring smooth interaction through the touchscreen display.

The GUI is structured with clear buttons, labels, entry fields, and navigation controls to improve usability and accessibility. The developed interface is illustrated in Figure 5.1 Python-Based GUI – Home Screen (Registration Selection).

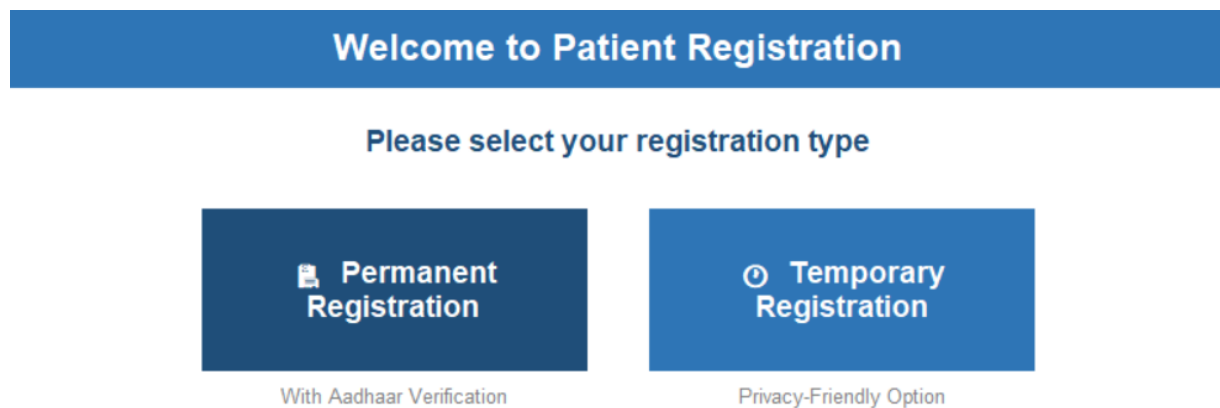


Figure 5.1 Python-Based GUI – Home Screen (Registration Selection)

The graphical user interface (GUI) includes multiple interactive screens to support the complete patient registration and preliminary diagnosis workflow. It provides a registration screen for entering patient details, a symptom input interface, real-time display of sensor readings, and an output screen showing the generated token along with department allocation results.

The GUI is designed with clear navigation controls, readable text, and a user-friendly layout to ensure accessibility for patients of different age groups and technical backgrounds. This interface plays a crucial role in enabling smooth and intuitive interaction between the user and the smart healthcare system. The detailed symptom entry and health monitoring interface is illustrated in Figure 5.2 GUI Screen – Symptom Input and Health Checkup Page.

Select Symptoms

Please select all applicable symptoms

☐ Fever	☐ Chest Pain	☐ Shortness of Breath
☐ Headache	☐ Abdominal Pain	☐ Joint Pain
☐ Ear Pain	☐ Eye Problem	☐ Skin Rash
☐ Urinary Problem	☐ Dental Pain	☐ Anxiety / Depression

Next – Health Check →
← Back

Figure 5.2 GUI Screen – Symptom Input and Health Checkup Page

5.2.4 Object-Oriented Programming

Object-Oriented Programming (OOP) is used to structure the software into reusable, modular, and easily manageable components, making the overall system more organized and scalable. In the proposed smart healthcare system, classes and objects are created to represent different functional modules such as sensor interfacing, graphical user interface, authentication, patient data handling, and decision processing.

This programming approach improves code organization by separating functionalities into dedicated classes, which enhances readability and reduces complexity. It also supports reusability of components, allowing the same modules to be extended or reused in future upgrades without rewriting the entire code. Furthermore, OOP simplifies debugging, testing, and maintenance, as errors can be isolated within specific classes or methods.

The modular structure also provides better scalability for future enhancements, such as adding cloud connectivity, new medical sensors, or AI-based diagnosis support. Hence, the use of OOP ensures that the software remains efficient, maintainable, and adaptable for long-term healthcare automation applications...

5.3 DATABASE MANAGEMENT

The proposed system uses a database module to securely store patient details, health parameters, and registration records generated during the automated workflow. A lightweight database such as SQLite is selected because of its simplicity, low memory requirement, and seamless compatibility with Python, making it highly suitable for Raspberry Pi-based embedded healthcare applications.

The database stores patient identification details, symptoms entered through the graphical user interface, sensor readings such as body temperature and heart rate, and the final department allocation results generated by the decision engine. In addition, token information and consultation timestamps can also be maintained for proper hospital queue management.

Efficient database management ensures fast data retrieval, reliable storage, and secure handling of sensitive patient information, which is essential for healthcare environments. It also helps preserve patient history for future reference, analytics, follow-up diagnosis, and system performance evaluation, thereby improving the intelligence and usefulness of the smart healthcare kiosk over time.

5.4 SENSOR CALIBRATION AND TESTING

Sensor calibration is essential to ensure accurate measurement of health parameters. The sensors used in the system are tested and calibrated under controlled conditions before deployment.

$$Emergency = \begin{cases} True, & T > 38.5^{\circ}C \\ True, & HR < 50 \\ True, & HR > 120 \\ False, & otherwise \end{cases} \dots\dots(1)$$

Error Calculation

$$\% \text{ Error} = \frac{|Measured - Actual|}{Actual} \times 100 \dots\dots\dots (2)$$

Calibration is an essential process to ensure that the sensors used in the proposed smart healthcare system provide accurate and reliable measurements. It involves comparing the sensor readings with standard medical or laboratory instruments, identifying any deviation, and then adjusting the sensor parameters to improve measurement accuracy.

The calibration process also includes testing the sensors under different environmental and operating conditions, such as varying temperature, humidity, and patient usage scenarios, to verify consistent performance. Proper calibration significantly improves the reliability, repeatability, and precision of diagnostic data, ensuring that the system generates trustworthy temperature, heart rate, and environmental readings for preliminary diagnosis and department allocation.

5.4.1 Temperature Sensor Results

The MLX90614 temperature sensor is tested by measuring body temperature under different conditions. The readings are compared with standard thermometer values to verify accuracy.

$$F = \frac{9}{5}C + 32 \dots\dots\dots(3)$$

Temperature in Celsius = C(°C)

Temperature in Fahrenheit = F(°F)

Table 5.1 Temperature Sensor Test Results

Sample No	Measured Temperature (°C)	Standard Temperature (°C)	Error (%)
1	36.5	36.6	0.27
2	36.8	36.9	0.27
3	37.0	37.1	0.27
4	36.7	36.8	0.27
5	36.9	37.0	0.27

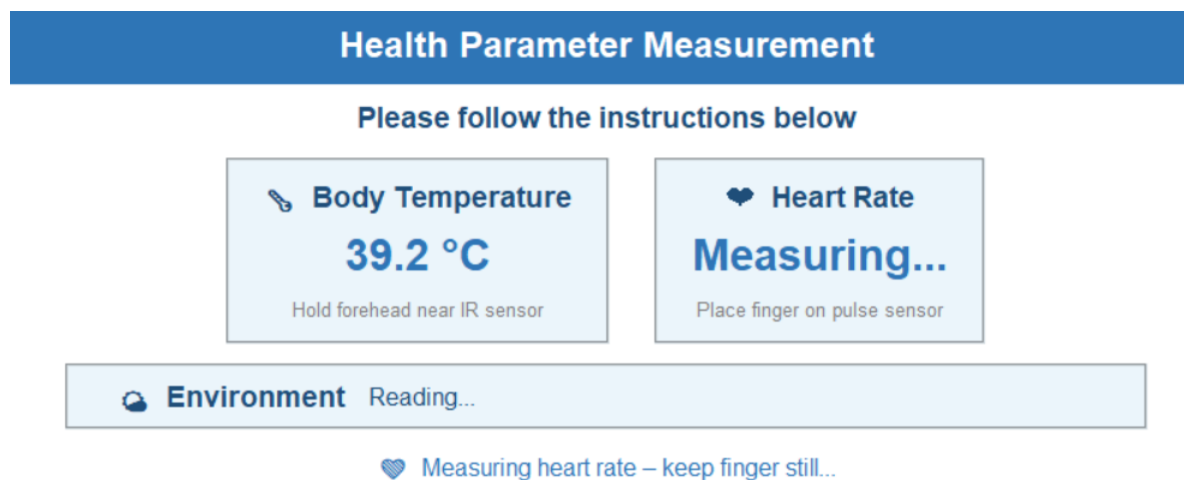


Figure 5.3 Sensor Output – Body Temperature Reading

The results indicate that the sensor provides accurate body temperature readings with minimal error, making it highly suitable for preliminary healthcare assessment and patient screening applications. The stable performance and quick response time of the sensor improves the reliability of the proposed smart healthcare kiosk during real-time operation. The measured output is illustrated in Figure 5.3 Sensor Output – Body Temperature Reading , while the corresponding experimental values are presented in Table 5.1 Temperature Sensor Test Results.

5.4.2 Heart Rate Sensor Results

The KY-039 pulse sensor is tested by measuring heart rate and comparing it with standard medical devices.

$$HR = 60/T \dots\dots\dots (4)$$

Heart Rate = HR (beats per minute, BPM)

Time interval between pulses = T(seconds)

Table 5.2 Heart Rate Sensor Test Results

Sample No	Measured Heart Rate (BPM)	Standard Value (BPM)	Error (%)
1	72	74	2.70
2	78	80	2.50
3	70	72	2.78
4	76	78	2.56
5	74	76	2.63

The results show that the pulse sensor provides reasonably accurate heart-rate readings, making it suitable for preliminary health assessment and basic patient screening within the proposed healthcare kiosk. The sensor demonstrates stable performance with acceptable response time, which supports reliable early-stage health monitoring. The detailed experimental observations and measured values are presented in Table 5.2 Heart Rate Sensor Test Results.

5.5 HARDWARE RESULTS AND DISCUSSIONS

The hardware model of the proposed system is developed by integrating all components, including Raspberry Pi, sensors, touchscreen display, and output modules. The system is tested under real-time conditions to evaluate its performance.

The hardware setup of the proposed smart healthcare kiosk operates successfully as expected, demonstrating reliable performance across all major functional stages. The system effectively performs patient registration through the touchscreen interface, enabling smooth and user-friendly data entry.

It also supports sensor-based measurement of body temperature and heart rate, where the MLX90614 and KY-039 modules capture real-time health parameters accurately. The acquired data is then processed and analyzed by the Raspberry Pi, which executes the embedded decision logic for symptom evaluation and patient routing.

Based on the analysis results, the system performs automatic department allocation and consultation token generation, ensuring faster patient movement and reduced waiting time. These successful hardware operations validate the practical feasibility and stability of the proposed healthcare automation system [15].

5.5.1 Hardware Model

The hardware model consists of all components assembled into a compact and functional system. Proper wiring and connections ensure stable operation. The system is designed to be user-friendly and suitable for deployment in hospital environments.

5.5.2 Working Status of the Hardware Model

The proposed system is tested with multiple users under different operating conditions to verify its performance, reliability, and response efficiency. The results obtained indicate that the system operates efficiently with minimal delays, even when handling repeated patient registrations and continuous sensor measurements.

The temperature and heart-rate sensor readings are observed to be accurate and consistent, while the software successfully processes patient inputs and generates

the required outputs such as department allocation and token details. During testing, several key observations were noted, including fast response time, accurate sensor measurements, smooth GUI operation, and reliable end-to-end system performance.

The overall performance evaluation of the hardware model clearly demonstrates the effectiveness of the proposed smart healthcare system in improving hospital registration, preliminary diagnosis, and patient routing processes, thereby validating its suitability for real-time healthcare environments [15].

CHAPTER 6

CONCLUSION & FUTURE EXTENSION

CHAPTER 6

CONCLUSION & FUTURE EXTENSION

6.1 CONCLUSION

The proposed IoT-based Automated Hospital Patient Registration and Preliminary Diagnostic System successfully address the major limitations of traditional healthcare registration methods, such as manual paperwork, long waiting time, repeated data entry, and inefficient patient routing. The system integrates modern technologies including embedded processing, IoT sensors, touchscreen interfacing, and software-based automation to provide an efficient, intelligent, and user-friendly solution for hospital patient management.

The implementation of a touchscreen-based self-registration interface enables patients to enter their details easily without manual staff assistance, thereby improving accessibility and reducing administrative workload. In addition, the integration of MLX90614 temperature sensing and KY-039 heart-rate monitoring allows real-time measurement of vital health parameters during the registration process itself. This eliminates the need for separate preliminary screening procedures and significantly reduces patient waiting time while improving operational speed.

The system also incorporates intelligent rule-based logic for symptom analysis, automated department allocation, and consultation token generation, ensuring that patients are directed to the appropriate medical department without unnecessary delay. The inclusion of digital authentication methods such as Register number verification improves data accuracy, prevents duplication of records, and enhances the reliability of patient information management. Overall, the proposed solution improves hospital workflow efficiency, reduces manual effort, minimizes congestion, and enhances the quality of healthcare service delivery.

The experimental test results demonstrate that the system performs reliably with accurate sensor readings, fast response time, and efficient data processing. Both the hardware and software components operate seamlessly together, making the system highly suitable for real-time deployment in modern healthcare environments. Thus, the proposed smart healthcare kiosk proves to be a practical, scalable, and effective solution for modernizing hospital registration and preliminary diagnostic processes.

6.2 FUTURE ENHANCEMENT

Although the proposed system performs effectively and demonstrates reliable automation of patient registration and preliminary diagnosis, there are several opportunities for further improvement and expansion to enhance its capabilities in real-world smart healthcare environments.

Future development can include integration with cloud platforms for secure remote data storage, backup, and authorized medical access from different hospital departments [2], [23]. The system can also be extended with advanced data analytics and AI-assisted processing to improve diagnostic accuracy and support predictive healthcare decision-making.

Additional enhancements may involve the integration of more biomedical sensors, such as blood pressure, SpO₂ (oxygen level), ECG, and glucose monitoring modules, enabling a more comprehensive preliminary health assessment [10], [21]. A dedicated mobile application for remote patient interaction, token tracking, and appointment status updates can further improve usability and accessibility. The user interface can also be upgraded with multi-language support to make the system easier to use for patients from different linguistic backgrounds.

From a system-level perspective, stronger security mechanisms, encrypted patient data storage, and secure authentication protocols can be incorporated to protect sensitive medical information and ensure privacy compliance [24]. Furthermore, seamless integration with hospital management systems (HMS), electronic health records (EHR), and doctor dashboards can enable smooth interdepartmental data sharing and workflow coordination. [25].

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APPENDICES I

PROGRAM SOURCE CODE

APPENDICES I

PROGRAM SOURCE CODE

File: main.py

```
import tkinter as tk

import threading

import sys

import os

import time

from tkinter import messagebox

sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), '..')))

from config import (WINDOW_WIDTH, WINDOW_HEIGHT,
FULLSCREEN, COLOR_WHITE, COLOR_DARK, COLOR_BLUE,
FONT_TITLE, FONT_HEADING, HOSPITAL_NAME, HOSPITAL_CITY)

from database import init_database

from services.admin_server import start_admin_dashboard

from sensors import read_temperature, read_heart_rate, read_environment

from sensors.spo2 import read_spo2

class HospitalKioskApp:

    """Concise representation of the Hospital Registration System."""

    def __init__(self, root):

        self.root = root

        self.root.title("Hospital Patient Registration Kiosk")
```

```

self.root.geometry(f'{WINDOW_WIDTH}x{WINDOW_HEIGHT}')

self.root.configure(bg=COLOR_WHITE)

if FULLSCREEN: self.root.attributes("-fullscreen", True)

self.name_var, self.age_var = tk.StringVar(), tk.StringVar()

self.vitals = {"temp": 0.0, "hr": 0, "spo2": 0, "env": {}}

init_database()

start_admin_dashboard()

self.show_welcome()

def clear(self):

    for widget in self.root.winfo_children(): widget.destroy()

def make_header(self, title):

    tk.Label(self.root, text=HOSPITAL_NAME, font=("Arial", 9, "bold"),
bg=COLOR_DARK, fg=COLOR_WHITE, pady=4).pack(fill=tk.X)

    tk.Label(self.root, text=title, font=FONT_TITLE, bg=COLOR_BLUE,
fg=COLOR_WHITE, pady=10).pack(fill=tk.X)

def make_footer(self, text=""):

    tk.Label(self.root, text=text or HOSPITAL_CITY, font=("Arial", 10),
bg=COLOR_DARK, fg=COLOR_WHITE, pady=3).pack(side=tk.BOTTOM,
fill=tk.X)

def show_welcome(self):

    self.clear()

    self.make_header("Patient Registration")

    card = tk.Frame(self.root, bg="#F0F0F0", bd=2, relief=tk.GROOVE)

```

```

card.pack(pady=40, padx=50, fill=tk.X)

tk.Label(card, text="Name:", font=FONT_HEADING,
bg="#F0F0F0").grid(row=0, column=0, pady=10, padx=10)

tk.Entry(card, textvariable=self.name_var,
font=FONT_HEADING).grid(row=0, column=1, pady=10, padx=10)

tk.Label(card, text="Age:", font=FONT_HEADING,
bg="#F0F0F0").grid(row=1, column=0, pady=10, padx=10)

tk.Entry(card, textvariable=self.age_var,
font=FONT_HEADING).grid(row=1, column=1, pady=10, padx=10)

tk.Button(self.root, text="Capture Vitals", font=FONT_HEADING,
bg=COLOR_BLUE, fg=COLOR_WHITE,
command=self.start_vitals_scan).pack(pady=20)

self.make_footer("Self-Service Patient Registration Kiosk")

def start_vitals_scan(self):

    if not self.name_var.get() or not self.age_var.get():

        messagebox.showwarning("Required", "Please enter Name and Age")

        return

    self.clear()

    self.make_header(f"Scanning {self.name_var.get()}...")

    self.status_lbl = tk.Label(self.root, text="Initializing Sensors...",
font=FONT_HEADING, bg=COLOR_WHITE)

    self.status_lbl.pack(pady=20)

    self.prog_lbl = tk.Label(self.root, text="● ● ●", font=("Arial", 32, "bold"),
fg=COLOR_BLUE, bg=COLOR_WHITE)

```

```

self.prog_lbl.pack(pady=10)

def animate(dots=0):

    if not hasattr(self, "status_lbl") or not self.status_lbl.winfo_exists():
return

    colors = ["#D1D5DB", "#9CA3AF", "#4B5563", "#2E75B6"]

    self.prog_lbl.config(fg=colors[dots % 4])

    self.root.after(400, lambda: animate(dots + 1))

animate()

def read_sensors():

    steps = [(" 🩺 Body Temperature", read_temperature, "temp"), (" ❤️ Heart
Rate", read_heart_rate, "hr"), (" 🩸 SpO2 Level", read_spo2, "spo2"), (" 🌫️
Environment", read_environment, "env")]

    for label, func, key in steps:

        self.root.after(0, lambda l=label:
self.status_lbl.config(text=f"Scanning: {l}"))

        time.sleep(2)

        self.vitals[key] = func()

        self.root.after(0, self.show_report)

    threading.Thread(target=read_sensors, daemon=True).start()

def show_report(self):

    self.clear()

    self.make_header("Final Patient Report")

    rep = tk.Frame(self.root, bg="#FFFFFF", bd=3, relief=tk.RAISED)

```

```

rep.pack(pady=30, padx=60, fill="both", expand=True)

tk.Label(rep, text="PATIENT SUMMARY", font=("Arial", 16, "bold"),
fg=COLOR_BLUE, bg="#FFFFFF").pack(pady=10)

details = [("Name", self.name_var.get()), ("Age", self.age_var.get()),
("Body Temp", f'{self.vitals['temp']} C'), ("Heart Rate", f'{self.vitals['hr']}
BPM'), ("Oxygen SpO2", f'{self.vitals['spo2']}%'), ("Humidity",
f'{self.vitals['env'].get('humidity', '--')}%'), ("Ambient Temp",
f'{self.vitals['env'].get('amb_temp', '--')} C')]

for label, val in details:

    f = tk.Frame(rep, bg="#FFFFFF")

    f.pack(fill="x", padx=40, pady=5)

    tk.Label(f, text=f'{label}:', font=("Arial", 12, "bold"), bg="#FFFFFF",
width=15, anchor="w").pack(side="left")

    tk.Label(f, text=val, font=("Arial", 12), bg="#FFFFFF",
anchor="w").pack(side="left")

tk.Button(self.root, text="Finish & New Registration",
font=FONT_HEADING, bg="#28a745", fg=COLOR_WHITE,
command=self.show_welcome).pack(pady=20)

self.make_footer("Thank you for using our service!")

if __name__ == "__main__":

    root = tk.Tk()

    app = HospitalKioskApp(root)

    root.mainloop()

```




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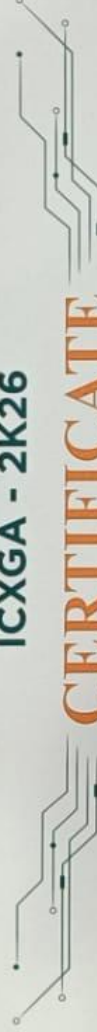
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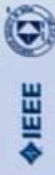


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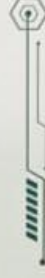


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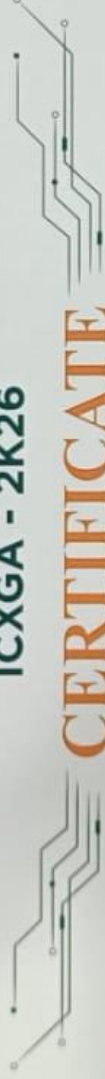
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FORM 1
THE PATENTS ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003
APPLICATION FOR GRANT OF PATENT
[See sections 7,54 & 135 and rule 20(1)]

(FOR OFFICE USE ONLY)

Application No.:
Filing Date:
Amount of Fee Paid:
CBR No.:
Signature:

1. APPLICANT(S):

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3. TITLE OF THE INVENTION: AN AUTOMATED HOSPITAL PATIENT REGISTRATION AND PRELIMINARY DIAGNOSTIC SYSTEM USING IOT-BASED SENSORS

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Sr.No.	Country	Application Number	Filing Date	Name of the Applicant	Tilte of the Invention
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6. PARTICULARS FOR FILING PATENT COOPERATION TREATY (PCT) NATIONAL PHASE APPLICATION:

International Application Number	International Filing Date as Allotted by the Receiving Office
PCT//	

7. PARTICULARS FOR FILING DIVISIONAL APPLICATION

Original (first) Application Number	Date of Filing of Original (first) Application
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8. PARTICULARS FOR FILING PATENT OF ADDITION:

Main Application / Patent Number:	Date of Filing of Main Application
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9. DECLARATIONS:

(i) Declaration by the inventor(s)

I/We ,KADHIRAVAN K ,AKASH KUMAR C M ,JAYAPRASANTH S ,SANJAI S ,DR. RAJESH KANNA R , is/are the true & first inventor(s) for this invention and declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) of the inventor(s):

(c) Name(s): KADHIRAVAN K ,AKASH KUMAR C M ,JAYAPRASANTH S ,SANJAI S ,DR. RAJESH KANNA R

(ii) Declaration by the applicant(s) in the convention country

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) :

(c) Name(s) of the singnatory: SREEJITH R

(iii) Declaration by the applicant(s)

- The Provisional specification relating to the invention is filed with this application.
- I am/We are, in possession of the above mentioned invention.
- There is no lawful ground of objection to the grant of the Patent to me/us.
- I am/We are, the assignee or legal representative to true first inventors.

10. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION:

Sr.	Document Description	FileName
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I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated hering are correct and I/We request that a patent may be granted to me/us for the said invention.

Dated this(Final Payment Date): -----

Signature:

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